

Curriculum Vitae

Praveen Jayakumar

PhD candidate, University of Toronto
praveen.jayakumar@mail.utoronto.ca
Website: praveen91299.github.io
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RESEARCH

Making quantum computers useful and reliable.

PUBLICATIONS

P. Jayakumar, T. Zeng, A. F. Izmaylov, "On the Feasibility of Exact Unitary Transformations for Many-body Hamiltonians." J. Chem. Theory Comput. 2026, 22, 8, 3916–3926, April 2026.

S. Patel, **P. Jayakumar**, R. Huang, T. Zeng, A. F. Izmaylov, "Quantum Seniority-based Subspace Expansion: Linear Combinations of Short-Circuit Unitary Transformations for Efficient Quantum Measurements." J. Chem. Theory Comput. 2026, 22, 8, 3937–3949, April 2026.

S. Patel, **P. Jayakumar**, T. C. Yen, A. F. Izmaylov, "Quantum Measurement for Quantum Chemistry on a Quantum Computer." Chem. Rev. 2025, 125, 16, 7490–7524, July 2025.

P. Jayakumar, P. J. Nadkarni, and S. S. Garani, "Efficient recursive encoders for quantum Reed-Muller codes towards Fault tolerance." arXiv:2405.14549, May 2024.

P. J. Nadkarni, **P. Jayakumar**, A. Behera, and S. S. Garani, "Entanglement-assisted Quantum Reed-Muller Tensor Product Codes." Quantum 8:1329, May 2024.

CONFERENCES

Poster titled "*Quantum Chemistry on a Quantum Computer*." presented at the Frontiers of Innovation conference 2026, University of Toronto. **poster contest - 2nd** place. March, 2026

Poster titled "*On the Feasibility of Exact Unitary Transformations for Many-body Hamiltonians*." presented at the Applied Quantum Computing Symposium 2025, University of Sherbrooke. November, 2025

Talk titled "*Conditions for Efficient Unitary Transformation of Many-Body Hamiltonians*" presented at the Symposium of Chemical Physics, University of Waterloo. November, 2025

Poster titled "*Reductions in Heisenberg transformed Hamiltonians for Distributed Quantum Chemistry*." presented at the Annual General Meeting of the Quantum Software Consortium. July, 2025

Talk titled "*Error mitigation via measurement groups: Making quantum computers to do chemistry*" presented at the Symposium of Chemical Physics, University of Waterloo. November, 2023

PREVIOUS RESEARCH PROJECTS	Fermionic reflection ansatz for efficient VQE methods Under the guidance of Prof. Artur F. Izmaylov. Novel variational ansatz construction. Published in JCTC.	July, 2021- May, 2022
	Quantum communication and error correction Under the guidance of Dr Shayan Srinivasa Garani ¹ – Studied entanglement properties of quantum RM codes and it's usecases on quantum networks and entanglement distribution/purification. arXiv:2405.14549 – Designed entanglement assisted codes, quantum tensor product codes from RM codes and their entanglement assisted versions that show positive catalytic rates. Presented, ITA 2023 and Published in Quantum	May, 2019- July, 2021
	iGEM 2019, Hardware team Design the hardware components of the project Synshine, IISc Bangalore.	December, 2018- November, 2019
CONTRIBUTED WORKSHOPS	Ion trapping for Ca⁺ ions Under the guidance of (late) Prof. Vasant Natarajan ² Constructing a linear Paul trap to trap singly ionized Calcium ions.	May, 2018- July, 2018
	<i>“Developing Effective Methodologies to Teach Quantum Information Science to Early-Stage Learners”</i> , IEEE Quantum Week 2021.	October, 2021
	<i>“Introduction to Quantum Computing”</i> , Quantum Communications Workshop 2021, IEEE-IISc.	September, 2021
	<i>“Introduction to Quantum Computing Algorithms”</i> , IEEE-IISc Student Seminar Series.	July, 2021
PARTICIPATED WORKSHOPS	IBM GSS 2021, IBM GSS 2020, STAQ summer school (online)	2020, 2021
	Radio Astronomy Workshop at ICTS, Bengaluru, India.	2018
	Duke TIP India Summer workshop on Cryptography by Duke University.	2012
EDUCATION	University of Toronto PhD candidate, under supervision of Prof. Artur Izmaylov ³ Department of Chemistry Research focus: Quantum computing algorithms for Quantum Chemistry Current grade: A	Fall, 2022 - Present
	Indian Institute of Science (IISc), Bengaluru, India 5-Year Bachelor of Science + Masters of Science with a major in Physics CGPA - 8.9 / 10.0 (Distinction) GRE (General): 326 /340 ToEFL: 103 /120	Fall, 2017 - Winter, 2022
	AMM Matriculation PUC I and II (equivalent to grades 11 and 12), Department of Government Examinations, TAMIL NADU. Class 12 : 96 %	Fall, 2015 - Winter, 2017

¹Department of Electronic Systems Engineering, IISc. shayang@iisc.ac.in

²Department of Physics, Indian Institute of Science

³Department of Physical and Environmental Sciences, University of Toronto Scarborough. artur.izmaylov@utoronto.ca

	AMM Matriculation Grades 1 - 10, Department of Government Examinations, TAMIL NADU. Class 10 : 97.2 %	Fall, 2005 - Winter, 2015
WORK EXPERIENCE	Xanadu Quantum Technologies Inc., Toronto, Canada. Summer Resident - Error correction researcher - GKP code and decoder design. Office of Communication, Indian Institute of Science, Bengaluru, India. Scientific journalism summer internship	May, 2024 - August, 2024 March, 2022 - July, 2022
TEACHING	Teaching assistant, CHM21B (Chemical Structure and Spectroscopy) Teaching assistant and lab demonstrator, PHYA11 Marker, CHM1478HS (Quantum Mechanics for Physical Chemists) Teaching assistant and lab demonstrator, PHYA10H Lab demonstrator, CHM135H TA, Digital signal processing (NPTEL)	Winter, 2025 Winter, 2024; Fall, 2025 Winter, 2024; Fall 2025 Fall, 2023; Fall, 2024 Fall, 2022 2020, 2021
RELEVANT COURSES	Graduate level COMPLETED Quantum computing, Quantum Mechanics for Quantum Chemists, Quantum information Theory, Quantum Optics, Linear Algebra and Optimization, Statistical Mechanics, Advanced Math methods, Information theory, Quantum mechanics, Advanced cryptography (Pseudo-randomness, randomness extractors and expanders), Quantum measurements, Theoretical foundations of Cryptography	
TECHNICAL SKILLS	<i>Programming languages</i> Python, L ^A T _E X, C, C++, Octave, MATLAB, R Quantum computing packages: Qiskit, Cirq, Tequila, PennyLane	
FELLOWSHIPS AND AWARDS	Kishore Vaigyanik Protsahan Yojana [KVPY-SA] <i>All India Rank : 94</i> Includes stipend and contingency Student of the year AMM School, Chennai, India	August, 2017- July, 2022 2017
EXTRA- CURRICULAR	Member, University of Toronto Toastmasters club Region 6, district 60, Division A, Area 11. Contributor and curator - quantumuniversal.org Executive member - IEEE IISc Communication Society student chapter.	May, 2023 - February, 2025 August, 2020 - Present April, 2021 - April, 2022

Quarks 2019 IISc UG's Annual magazine - design coordinator.	June, 2019 - March, 2020
Notebook drive, IISc Teach basic English, computer skills and science to underprivileged children.	August, 2018 - January, 2020
Pravega - Coordinator - Corporate relations Managing corporate communications and sponsorship for Pravega 2019, IISc's Annual Science, technical and cultural fest.	January, 2018 - January, 2019
Undergraduate engineering club, IISc - Coordinator Conduct and arrange seminars on engineering topics.	June, 2018 - June, 2019
Student Council (School) - Head	July, 2016 - April, 2017
